PAUL-ANDREI SIPOS

Python Developer | Automation Engineer | Backend Engineer | Test Engineer

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with 5+ years of experience in Python development, backend systems, and large-scale test automation within the automotive industry. Specialized in building automation tools, integrating REST APIs, and developing scalable backend solutions that eliminate manual work and improve engineering productivity.

At Elektrobit Automotive, developed Python automation frameworks that automated approximately 70% of manual testing workflows (~1500+ test cases) by extracting test data from Polarion via REST APIs and converting them into executable tests for internal platforms such as ATP and EB ITA. Additionally built internal developer tools and UI utilities for sending UDP signals to embedded targets, simplifying debugging and manual test execution.

Known for proactively identifying repetitive engineering processes and designing automation systems and developer tooling that streamline workflows, increase testing efficiency, and enable teams to focus on higher-value engineering work.

TECHNICAL STACK

Languages: Python, C/C++, C#, JavaScript

Backend: REST APIs, API Integrations, Microservices, Polarion API, GitLab API

Automation: Python Automation, Test Automation, Workflow Automation, scripting

Databases: PostgreSQL, SQL data processing, Grafana

DevOps: Git, GitHub, Gerrit, GitLab, Jenkins, CI/CD Pipelines

Networking: UDP Communication, Signal Injection

Methodologies: Agile, Scrum, Continuous Integration

INTERNSHIPS & EARLY EXPERIENCE

- Built web applications using Node.js and Angular on IBM Cloud.
- Developed interactive applications in Unity using C#.
- Worked with SQL databases and data manipulation systems.

PROFESSIONAL EXPERIENCE

Software Engineer - Elektrobit Automotive Romania

Nov 2021 – Present

- Develop backend and automation tools for automotive software testing environments.
- Built Python automation tools that automated ~70% of manual testing (~1500+ test cases).
- Extracted test cases and metadata from Polarion via REST APIs and converted them into executable tests for ATP and EB ITA platforms.
- Designed automation frameworks that enabled faster regression testing and more reliable validation.
- Developed UI tools for sending UDP signals to embedded targets to simplify testing and debugging.
- Integrated systems using REST APIs including Polarion and GitLab.
- Contributed to CI/CD automation using Jenkins and GitLab.
- Worked in Agile Scrum teams collaborating with developers, testers and system engineers.

EDUCATION

Faculty of Electrical Engineering and Computer Science - 2018-2022

Specialized in systems engineering, robotics, digital electronics, object-oriented programming, SQL and MATLAB.

LANGUAGES

- English - Advanced
- French - Elementary
- Spanish - Elementary

ADDITIONAL STRENGTHS

- Strong automation mindset focused on eliminating repetitive engineering work.
- Experience working with complex automotive software systems.
- Ability to design developer productivity tools and internal automation platforms.
- Strong problem-solving and cross-team collaboration skills.
- Use of AI tools to assist with code generation, debugging, and automation to improve development efficiency.